Schistosoma

**Background document for the
WHO Guidelines for drinking-water quality and
the WHO Guidelines on sanitation and health**

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Highlights

- There is strong evidence linking transmission of *Schistosoma* to inadequate sanitation, poor hygiene and lack of access to safe drinking-water. However, there is no evidence of transmission through ingestion of drinking-water. The provision of an adequate water supply contributes to disease prevention by reducing the use of infested water for domestic purposes such as bathing and washing clothes.
- *Schistosoma* are blood flukes, with most human infections caused by *S. mansoni*, *S. haematobium* and *S. japonicum*. Humans are the definitive vertebrate host, but a wide range of animals have been shown to be infected with human schistosomes. Hybridization between human and animal schistosomes represent a potential concern for zoonotic transmission.
- *Schistosoma* are transmitted by skin contact through domestic (e.g bathing, laundry), occupational (e.g. fishing) and recreational use of contaminated water. Transmission involves a complex two-host life cycle starting with contamination of freshwater with eggs from infected humans (or animals) via faeces or urine, reproduction in an intermediate snail host, and release of cercariae that infect the vertebrate host by penetrating the skin. Inside hosts, they develop into adult flukes, mate and produce eggs. Person-to-person transmission does not occur.
- Schistosomiasis involves acute and chronic symptoms that present in two main forms: intestinal (*S. mansoni*, *S. japonicum*, *S. mekongi*, *S. intercalatum*) and urogenital schistosomiasis (*S. haematobium*). Symptoms are typically based on immune responses to eggs and are influenced by amount and location of eggs in the body.
- *Schistosoma* are endemic in tropical and subtropical regions. Geographical distribution of individual species varies as each has a specific range of snail hosts. Most infections occur in low-income populations with inadequate access to safe drinking-water and effective sanitation. People of all ages are susceptible to infection, but school-age children are particularly vulnerable due to increased swimming and fishing in contaminated water and inadequate hygiene.
- Schistosome eggs in urine and faeces are detected via microscopic examination and molecular methods. Qualitative methods for the detection of cercariae in standing water are available.
- In endemic areas, host snails are susceptible to invasion by schistosome eggs in faeces and urine. Once released from the host snail, cercariae can remain infective for 1 to 3 days in freshwater.
- Systematic approaches that identify and control faecal and urine contamination of surface waters, snail control programmes and provision of safe drinking-water to reduce exposure to contaminated water through domestic use is needed to manage risks from *Schistosoma*.
- Schistosome eggs should be removed by conventional wastewater and drinking-water treatment processes through size exclusion (e.g. sand filtration, membranes) or settling processes (e.g. sedimentation basins, stabilization ponds). Disinfection processes, including chlorination and UV light, are effective for inactivating cercariae.
- Due to the complex life cycle and environmental survival of *Schistosoma*, *E. coli* (or, alternatively, thermotolerant coliforms) is not a suitable indicator for the possible presence or absence of *Schistosoma* in drinking-water.

1 General description

Schistosoma (commonly known as blood flukes) is a genus of trematodes. There are 23 species of *Schistosoma*. Most human infections are caused by *S. haematobium, S. mansoni*, and *S. japonicum*; but other species can infect humans, namely *S. mekongi*, *S. guineensis* and *S. intercalatum*. Natural hybridizations are known within and between animal and human schistosome species. Hybridizations between animal and human parasites are a public health concern due to potential for novel zoonotic transmissions (Leger & Webster, 2017).

Schistosoma flukes include male and female specimens. Human schistosomes can be 7 mm to 20 mm long and 0.3 mm to 0.6 mm wide. Male flukes are shorter and thicker than the females. After mating, the females produce eggs that are oval-shaped and up to 100 mm by 30 mm to 50 mm. The infective cercariae have pear-shaped heads and forked tails and are 400–600 µm in length. Humans are the definitive hosts of human schistosomes, but 113 animal species have been reported to be naturally infected with human schistosomes and should be considered as potential reservoirs (Boissier et al., 2019).

2 Human health effects

Clinical manifestations of schistosomiasis, also known as bilharzia, vary by species (Boissier et al., 2019). General symptoms include abdominal pain, diarrhoea, and blood loss in urine or faeces. Anaemia and malnutrition can be associated with the disease with potential long-term effects on growth and cognitive development in children. Symptoms of schistosomiasis typically result from the immune response to eggs in tissues, and not from the flukes themselves. Therefore, the symptoms depend on the amount and location of eggs in the body.

There are two main forms of schistosomiasis. Intestinal schistosomiasis (caused by *S. mansoni*, *S. japonicum*, *S. intercalatum*, *S. guineensis* and *S. mekongi)* produces abdominal pains, diarrhoea, blood in stools and enlargement of the liver and spleen. Life-threatening complications arising from chronic infections include liver fibrosis and portal hypertension. Urogenital schistosomiasis (caused by *S. haematobium)* typically manifests as presence of blood in urine. Long-term urogenital schistosomiasis can cause bladder cancer and renal failure. Urogenital schistosomiasis can also cause genital lesions in men and women. In women, potential complications include miscarriages, ectopic pregnancy and infertility.

In some people, acute schistosomiasis (Katayama fever), including fever, rash, chills, enlarged liver and spleen, muscle pains and cough, can begin within 1 to 2 months post infection, particularly with *S. japonicum*, and is triggered by initial egg deposition by maturing schistosomes. Rarely, eggs are found in the brain or spinal cord and can cause cerebral symptoms, such as seizures and paralysis.

If cercariae of non-human infectious schistosomes penetrate the human skin, they can cause an inflammatory response, especially in hosts that have been exposed previously. Papular rash, known as schistosome cercarial dermatitis (or more commonly swimmer's itch), can result at points of penetration of cercariae. The cercariae of avian schistosomes are responsible for most cases of this type of dermatitis.

3 Epidemiology

Schistosomiasis affects over 251 million people worldwide (WHO, 2023a). Global distribution of the disease varies by causative species, since each has a specific range of suitable snail hosts: *S. mansoni* is found in Africa, the Middle East, the Caribbean, Brazil, Venezuela and Suriname; *S. haematobium* is found in Africa, the Middle East and the island of Corsica, France; *S. japonicum* is limited to China, Indonesia and the Philippines; *S. mekongi* is found in several districts in Lao People's Democratic Republic and Cambodia and *S. guineensis* and *S. intercalatum* are found in rainforest areas of central Africa (WHO, 2023a).

The distribution of schistosomes is largely dependent on the availability of freshwater habitats for the snail host, such as ponds, lakes and slow-flowing streams. Water resource development projects, including dam construction, the expansion of irrigation systems, and agricultural water infrastructure, have been identified as potential sources of elevated and shifting rates of schistosomiasis as a result of the creation of new habitats for freshwater snails (Steinmann et al., 2006).

Most infections occur in poor communities with lack of access to safe drinking-water and inadequate sanitation. In endemic areas, incidence varies spatially and seasonally with distance to water bodies, rainfall, temperature, and irrigation practices that influence exposure and the snail host. Prevalence of infection is linked to the proportion of a population having contact with contaminated surface water as well as frequency and duration of contact (Grimes et al., 2015). Exposure occurs through domestic (e.g. bathing, laundry), occupational (e.g. fishing) or recreational activities involving water contact. Use of untreated faecal sludge and wastewater in agriculture has been associated with increased risk of schistosome infections (Carlton et al., 2015; Furhimann et al., 2016). People of all ages are susceptible to infection, but school-age children are particularly vulnerable due to increased swimming and fishing in contaminated water and inadequate hygiene.

4 Transmission

Humans typically become infected by skin contact with contaminated water. The life cycle of schistosomiasis involves an intermediate snail host and a definitive vertebrate host, which may be human and/or animal, depending on the species. Schistosome eggs are shed in faeces (*S. mansoni*, *S. japonicum*, *S. intercalatum*, *S. guineensis* and *S. mekongi)* or urine (*S. haematobium*) of infected individuals. Once in contact with fresh water, the eggs hatch into free-swimming miracidia, which follow light and chemical stimuli to find and invade snail hosts. Within the snail host, miracidia undergo asexual reproduction. After several weeks, the infected snails start to release thousands of cercariae into the water. Cercariae find and infect definitive hosts by penetrating through the skin. Once in the human tissue, the cercariae transform into schistosomules, which migrate to the lungs through the circulatory system and develop into adult flukes in the mesenteric veins of the intestine or the bladder, depending on the species. Once the flukes mature, they mate and produce eggs. The entire life cycle from exposure to laying eggs requires about 3 to 4 months.

One cercaria is enough to cause infection, but both female and male flukes are required to produce eggs. The amount of cercariae produced per snail depends on the snail host, the environment and the time of shedding from the snail which coincides with the time where encounter with the definitive hosts is most probable (e.g. shedding in the morning when the definitive host is human, night when the definitive host is rodents) (Mouahid et al., 2012).

Schistosomes live for an average of 3 to 10 years in their human hosts, but as long as 40 years in some cases (Colley, 2014). Schistosomes can produce eggs for 3 to 8 years thus humans may shed eggs for years, if not decades without reinfection (Webber, 2005). Female flukes can produce eggs ranging from hundreds (*S. haematobium, S. mansoni)* to thousands (*S. japonicum)* per day. Egg output for *S. haematobium* and *S. mansoni* varies depending on age and is highest in 10 to14 year old children with means of about 5 *S. haematobium* eggs/10 mL of urine and 65 *S. mansoni* eggs/g of stools (Colley et al., 2014). *S. haematobium* eggs are thought to enter freshwater bodies through direct urination, often by children bathing and swimming in water (Jordan, 1978). *S. mansoni* eggs are thought to enter through open defecation along the shore as well as anal washing in freshwater bodies (Chandiwana, 1986; Sow et al., 2008). However, more than half of the eggs are trapped in tissues and not excreted in urine or faeces.

There is some evidence of increasing age-associated immunity in endemic populations (Mitchell et al., 2012).

5 Detection methods

Schistosome eggs in urine and faeces are detected via microscopic examination and molecular methods. Qualitative methods for the detection of cercariae in standing water include differential filtration, use of sieves, and cercarial attractants with unsaturated fatty acids (these methods have not been improved for moving water) (Boissier et al., 2019). Infected snails can be detected with a binocular microscope when shell transparency permits or by collecting and isolating individual snails and exposing them to two consecutive periods of 12 hours light and 12 hours dark and inspecting the water for the cercariae. However, neither of these methods guarantee the identification of cercariae as snails may be in the prepatent period of infection (i.e. infected but not shedding cercariae) (Boissier et al., 2019). Molecular assays have been developed to detect snail schistosomiasis for *S. mansoni and S. haematobium* (Abbasi, 2010).

6 Source and occurrence

6.1 Faecal pollution sources

No reports were identified on occurrence of schistosome eggs in wastewater or faecal sludge. Schistosome eggs can survive days to weeks outside water but require contact with water to hatch into miracidia (Maldonado et al., 1946). Eggs deposited by the shore or on riverbanks (e.g. through open defecation) may still hatch into miracidia if washed into the water body by rain (Jordan, 1978). Miracidia can only survive a few days in the absence of a suitable host snail (Anderson et al., 1982).

6.2 Surface water

No reports were identified on the occurrence of schistosome eggs in surface water. In endemic areas surface water containing host snails is susceptible to contamination by schistosome eggs in faeces and urine. Once released from the host snail, cercariae can remain infective to humans for 1 to 3 days in freshwater (Lawson & Wilson, 1980). Data on the occurrence of cercariae are limited. In one study concentrations in standing water ranged from 52 cercariae/10 L to 190 cercariae /10 L (Theron, 1979). Neither schistosome cercariae nor the intermediate host snails can survive in seawater (Oyerinde & Jaji, 1986).

6.3 Groundwater

Schistosome eggs cannot survive and complete their life cycle in groundwater (Boissier et al., 2019). Similarly, it is unlikely that cercariae could contaminate groundwater.

6.4 Soil and crops

Schistosome eggs can survive for extended periods in soil and on crops but must be washed into surface water to complete their life cycle (Boissier et al., 2019). It is possible that miracidia could be present on soils and crops irrigated with contaminated water but cannot survive in the absence of a suitable host.

7 Significance of sanitation

Safe sanitation systems that effectively separate human excreta (faeces and urine) from the environment and prevent excreta from contaminating snail-infested waters are an important barrier against contamination of water with *Schistosoma* eggs excreted by infected humans (Grimes et al., 2015). Management of animal faeces and urine and prevention of animal contact with surface water are also required as some schistosomes have animal reservoirs which contribute to human infection (e.g. *S. japonicum* in cattle, water buffalo, dogs, rats, etc.) (Rudge et al., 2013) and there are emerging animal livestock/human hybrids, such as, *S. haematobium* with *S. bovis,* as well as livestock *Schistosoma* hybrids infecting humans (Leger et al., 2016).

8 Significance in drinking-water

Although most infections occur in poor communities without access to safe drinking-water, there is no evidence linking transmission of schistosomiasis with ingestion of drinking-water. Readily available safe drinking-water contributes to disease prevention by reducing use of infested water for domestic purposes, e.g. washing clothes, bathing (Grimes et al., 2015).

9 Prevention and management

The control of schistosomiasis typically relies on periodic administration of the anti-parasitic drug praziquantel, primarily to school-aged children (Webster et al., 2014). However, this strategy alone is insufficient to interrupt transmission in the long term. Thus the role of preventative measures at the individual and community level are essential (Liang et al., 2007; Grimes et al., 2015; Zhang et al., 2022). Safe sanitation, access to safe drinking-water and hygiene are key measures.

Systematic approaches that identify and control faecal and urine contamination of surface water and provision of safe drinking-water to reduce exposure from domestic use of contaminated water is needed to manage risks from *Schistosoma*. Treatment processes based on size exclusion (e.g. filtration, membranes) or settling in primary wastewater treatment processes are effective at removing schistostome eggs. Disinfection processes, including chlorination and UV light, are also effective for inactivating or significantly damaging cercaria (Braun et al., 2018; 2020).

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Schistosome eggs and miracidia have been found to survive for relatively short periods of time in pit latrines, waste stabilization ponds and anaerobic biogas digesters (Kawata & Kruse, 1966; Remais et al., 2009). Safe containment (preventing direct leakage of pit contents into water bodies) and safe transport and treatment of faecal waste before disposal or end-use (e.g. no dumping of sludge into rivers) reduce the risk of contaminating snail-infested water bodies. This, in turn, reduces snail infections and the resulting cercaria production and hence lowers the risk of human infection.. Schistosome eggs, like other helminth eggs, are expected to be effectively removed by conventional wastewater treatment processes through size exclusion (e.g. sand filtration, membranes) (Braun et al., 2018) or settling processes (e.g. sedimentation basins, stabilization ponds). Systems involving primary and secondary sewage treatment provided 2.3 $\log_{10}$ removal of *S. mansoni* eggs (Rowan, 1964).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023b), control measures to manage potential risks from *Schistosoma* should include protection of source waters from human and animal faeces and urine, snail control programmes, application of adequate water treatment processes and protection of storage and distribution systems from external contamination. Schistosome cercariae can be removed or inactivated from water by filtration, heat treatment (Braun et al., 2018) or by disinfection. Although cercariae are moderately resistant to chlorination,.[1] doses and contact times typically used in drinking-water treatment are effective for inactivating cercariae A Ct (the product of disinfectant concentration and contact time) value of 26 mg.min/L has been shown to achieve 2 $\log_{10}$ inactivation of *S. mansoni* cercariae at pH 7.5 and 20 °C (Braun et al., 2020). UV is also effective at inactivating schistosome cercariae; doses between 3 and 60 mJ/cm^2 can significantly damage or inactivate cercariae (Braun et al 2018).

Due to the complex life cycle and environmental survival of *Schistosoma*, *E. coli* (or, alternatively, thermotolerant coliforms) is not a suitable indicator for the possible presence or absence of *Schistosoma* in drinking-water.

[1] Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct_{99} of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

10 ICD-11 classification

XN78L	Schistosoma
XN86N	Schistosoma haematobium
XN9FK	Schistosoma intercalatum
XN1ZJ	Schistosoma japonicum
XN8HD	Schistosoma mansoni
XN9T7	Schistosoma mekongi
1F86	Schistosomiasis
1F86.0	due to Schistosoma haematobium
1F86.1	due to Schistosoma mansoni
1F86.2	due to Schistosoma japonicum
1F86.4	Cercarial dermatitis

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book "*Sanitation and Disease in the 21st Century",* an update of the seminal book by Feachem et al. 1983 *"Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*" and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.